FORENSIC INTELLIGENCE AUDIT

Scenario B: Hidden Gems & Underrated Products Investigation

A Step-by-Step Practical Guide to What We Built, How We Analyzed It, and How to Recover These Listings

Step 1: Problem Definition & The Rating Discrepancy

What is Scenario B? It represents *Underrated Hidden Gems*: products displaying a poor public star rating (⭐ ≤ 3.0 / 5.0), but containing overwhelmingly positive customer reviews (🟢 ≥ 65% to 100% positive text sentiment). Shoppers avoid them thinking they are poor quality, but real buyers genuinely like them.

ANALYZED CATALOG	FLAGGED GEMS	AVERAGE DISPLAY STARS	UNDER-REVIEWED BIAS
15,641	**174**	⭐ **2.98**	**98.9%**
Total Products	Scenario B Items (1.11%)	Artificially Low Score	Sample Size Truncation

Step 2: Diagnostic Forensics & Why Ratings Tanked

Why did good products get a bad score? We analyzed review variance, star distributions, and buyer verification to diagnose the exact reason why these ratings collapsed:

Diagnostic Verdict	Product Count	Share (%)	Root Explanation
Sample Size Truncation	172	98.9%	Only 1 to 3 total reviews; 1 bad review dragged average down
Suspected Review Bombing	1	0.6%	An unverified 1-star account destroyed verified 5-star ratings
Polarized Opinions	1	0.6%	Split reactions (High variance between 1-star and 5-star reviews)

Key Insight: 98.9% of Scenario B products are not defective. They are simply buried by mathematical sample size bias rather than genuine customer dissatisfaction.

Step 3: Case Studies & Isolating the "Rating Killers"

Our engine analyzed individual review pairs to uncover how a single complaint ruined an otherwise loved product:

• **Case #1 (Product B0BDFT8DZR - Ice Maker):** Display rating is **3.0 Stars** with **75% positive share**. Happy buyers praised it as *"The good ice, mostly happy"*, but one review stating *"Smaller ice bucket than expected"* cut the score in half.

• **Case #2 (Product B0000GS2OG - Top Loader Washer):** An unverified buyer (`Verified=False`) left a 1-star review (*"Worst product I've EVER purchased"*), which dragged down legitimate verified praise and lowered the display score to 3.0.

Step 4: The Recovery Engine & Opportunity Math

How many reviews to fix these products? Using the formula `X = (Target × Total - Current Sum) / (5 - Target)`, we computed the exact number of subsequent 5-star reviews needed to lift each product back to **4.0+ Stars**:

Product ASIN	Current Stars	Positive Text %	Total Reviews	Reviews to Reach 4.0★
B01DKC2GQ0	3.00 / 5.0	100.0%	2	**Only 2 reviews**
B00Q9EJ45S	3.00 / 5.0	100.0%	2	**Only 2 reviews**
B0C8TR1SWY	3.00 / 5.0	100.0%	2	**Only 2 reviews**
B095RGB1WT	3.00 / 5.0	100.0%	1	**Only 1 review**
B09PL5HHQJ	3.00 / 5.0	100.0%	1	**Only 1 review**

Step 5: Business Recovery Action Plan
Unlike Scenario A (which requires supplier penalties and manufacturing fixes), Scenario B presents pure growth opportunity. Merchandising teams can send automated post-purchase review requests to verified buyers. Adding just 1 to 2 positive reviews instantly rescues 172 high-quality products back into Amazon search recommendations.